

Администрирование сетевых подсистем

Настройка безопасного удалённого доступа по протоколу SSH

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Цели и задачи работы

Цель лабораторной работы

Приобрести практические навыки настройки безопасного удалённого доступа к серверу с помощью SSH и средств контроля (sshd_config, firewall, SELinux, ключевая аутентификация, туннели).

Выполнение лабораторной работы

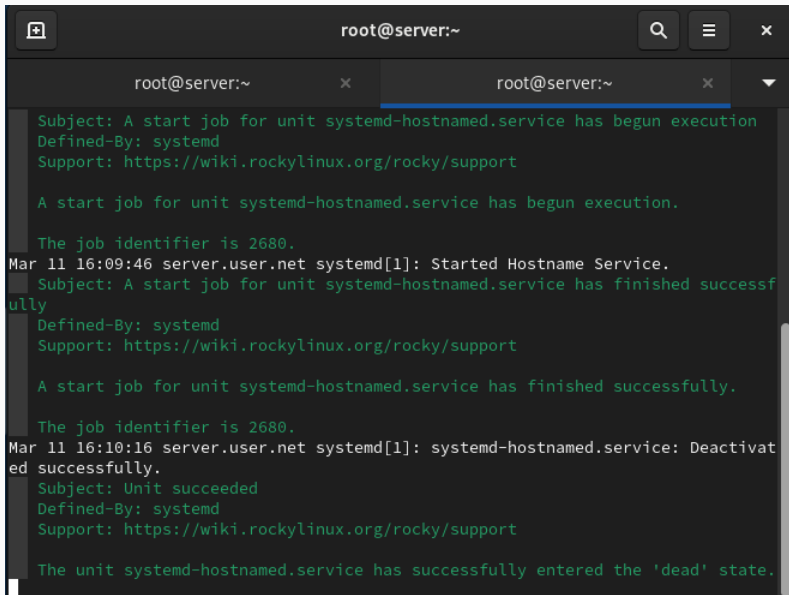
Мониторинг событий е попытка входа root

A terminal window with a dark background and light-colored text. The text shows a sequence of password change attempts for the root user. The first attempt fails with the message "BAD PASSWORD: The password is shorter than 8 characters". The user is prompted to "Retype new password:". The second attempt is successful, with the message "passwd: all authentication tokens updated successfully.". The user then enters the command "[root@server ~]# passwd root". The terminal shows the process of changing the password for the root user, including prompts for the new password and confirmation. The second attempt fails with the same error message. The third attempt is successful, and the terminal shows the command "[root@server ~]#" again.

```
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server ~]# passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@server ~]#
```

Рис. 1: Мониторинг событий е попытка входа root

Ошибки аутентификации root em журнале



A terminal window titled 'root@server:~' with search, menu, and close icons. It displays two tabs, both labeled 'root@server:~'. The active tab shows the following log output:

```
Subject: A start job for unit systemd-hostnamed.service has begun execution
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit systemd-hostnamed.service has begun execution.

The job identifier is 2680.
Mar 11 16:09:46 server.user.net systemd[1]: Started Hostname Service.
Subject: A start job for unit systemd-hostnamed.service has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit systemd-hostnamed.service has finished successfully.

The job identifier is 2680.
Mar 11 16:10:16 server.user.net systemd[1]: systemd-hostnamed.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit systemd-hostnamed.service has successfully entered the 'dead' state.
```

Запрет удалённого входа para root

```

31 #KeyExchange default none
32
33 # Logging
34 #SyslogFacility AUTH
35 #LogLevel INFO
36
37 # Authentication:
38
39 #LoginGraceTime 2m
40 PermitRootLogin no
41 #StrictModes yes
42 #MaxAuthTries 6
43 #MaxSessions 10
44
45 #PubkeyAuthentication yes
46
47 # The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
48 # but this is overridden so installations will only check .ssh/authorized_keys
49 AuthorizedKeysFile .ssh/authorized_keys
50
51 #AuthorizedPrincipalsFile none
52
53 #AuthorizedKeysCommand none
54 #AuthorizedKeysCommandUser nobody
55
56 # For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
57 #HostbasedAuthentication no
58 # Change to yes if you don't trust ~/.ssh/known_hosts for
59 # HostbasedAuthentication
60 #IgnoreUserKnownHosts no
61 # Don't read the user's ~/.rhosts and ~/.shosts files
62 #IgnoreRhosts yes
63
64 # To disable tunneled clear text passwords, change to no here!
65 #PasswordAuthentication yes
66 #PermitEmptyPasswords no
67
68 # Change to no to disable s/key passwords

```

Saving file "/etc/ssh/sshd_config"...

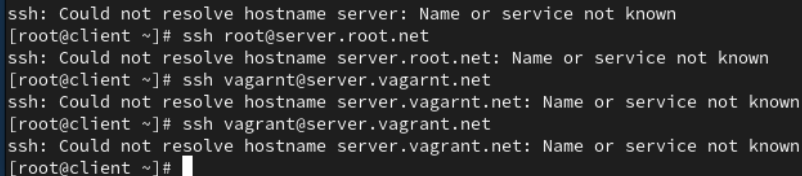
Plain Text ▾ Tab Width: 8 ▾ Ln 44, Col 1 ▾ INS

Результат после запрета root

```
[root@client ~]# ssh root@server.vagrant.net
ssh: Could not resolve hostname server.vagrant.net: Name or service not known
[root@client ~]# ssh root@server.root.net
ssh: Could not resolve hostname server.root.net: Name or service not known
[root@client ~]# ssh root@server@root.net
ssh: Could not resolve hostname root.net: Name or service not known
[root@client ~]# ssh root@server@vagrant.net
ssh: Could not resolve hostname vagrant.net: Name or service not known
[root@client ~]# ssh root@server
ssh: Could not resolve hostname server: Name or service not known
[root@client ~]# sudo -i
[root@client ~]# cd ../exit
-bash: cd: ../exit: No such file or directory
[root@client ~]# exit
logout
[root@client ~]# ssh root@server
ssh: Could not resolve hostname server: Name or service not known
[root@client ~]# ssh root@server.root.net
ssh: Could not resolve hostname server.root.net: Name or service not known
[root@client ~]#
```

Рис. 4: Отказ ем доступе root por SSH

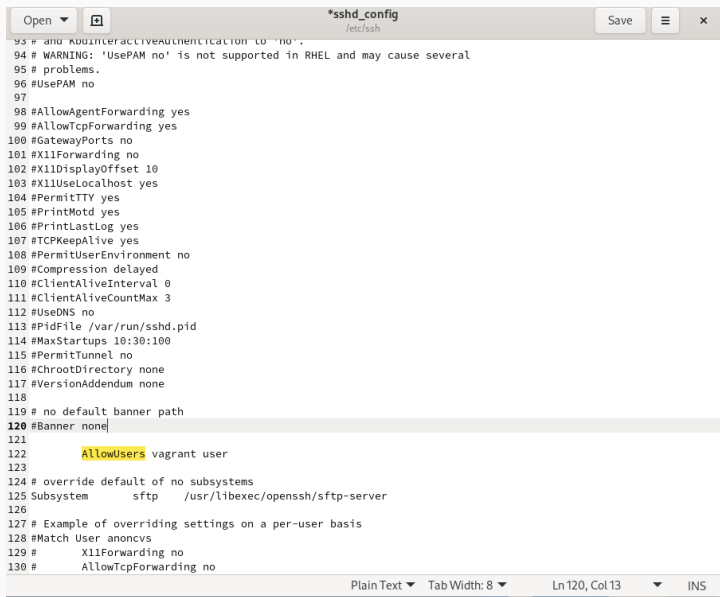
Успешное подключение пользователя akabrikosov

A terminal window with a dark background and light gray text. It shows four failed SSH connection attempts from a client to a server. Each attempt results in an error message: 'ssh: Could not resolve hostname server: Name or service not known' or 'ssh: Could not resolve hostname server.root.net: Name or service not known'. The user is root@client ~. The attempts are for root@server.root.net, vagrant@server.vagarnt.net, and vagrant@server.vagrant.net. The terminal ends with a prompt [root@client ~]# and a cursor.

```
ssh: Could not resolve hostname server: Name or service not known
[root@client ~]# ssh root@server.root.net
ssh: Could not resolve hostname server.root.net: Name or service not known
[root@client ~]# ssh vagarnt@server.vagarnt.net
ssh: Could not resolve hostname server.vagarnt.net: Name or service not known
[root@client ~]# ssh vagrant@server.vagrant.net
ssh: Could not resolve hostname server.vagrant.net: Name or service not known
[root@client ~]#
```

Рис. 5: Подключение пользователя akabrikosov

Ограничение списка пользователей AllowUsers

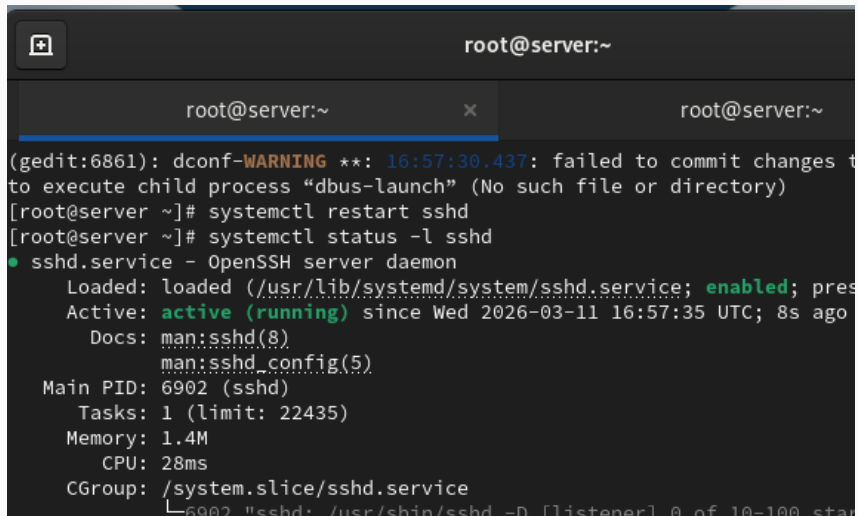


The image shows a text editor window titled '*sshd_config /etc/ssh'. The editor contains the configuration for the SSH daemon. The 'AllowUsers' directive is highlighted in yellow on line 122, showing 'vagrant user'. The status bar at the bottom indicates 'Plain Text', 'Tab Width: 8', and the cursor is at 'Ln 120, Col 13'.

```
Open [icon] *sshd_config /etc/ssh Save [icon] x
93 # and authentication to "no".
94 # WARNING: 'UsePAM no' is not supported in RHEL and may cause several
95 # problems.
96 #UsePAM no
97
98 #AllowAgentForwarding yes
99 #AllowTcpForwarding yes
100 #GatewayPorts no
101 #X11Forwarding no
102 #X11DisplayOffset 10
103 #X11UseLocalhost yes
104 #PermitTTY yes
105 #PrintMotd yes
106 #PrintLastLog yes
107 #TCPKeepAlive yes
108 #PermitUserEnvironment no
109 #Compression delayed
110 #ClientAliveInterval 0
111 #ClientAliveCountMax 3
112 #UseDNS no
113 #PidFile /var/run/sshd.pid
114 #MaxStartups 10:30:100
115 #PermitTunnel no
116 #ChrootDirectory none
117 #VersionAddendum none
118
119 # no default banner path
120 #Banner none
121
122     AllowUsers vagrant user
123
124 # override default of no subsystems
125 Subsystem      sftp    /usr/libexec/openssh/sftp-server
126
127 # Example of overriding settings on a per-user basis
128 #Match User anoncvs
129 #     X11Forwarding no
130 #     AllowTcpForwarding no
```

Plain Text ▾ Tab Width: 8 ▾ Ln 120, Col 13 ▾ INS

Отказ em доступе пользователю вне AllowUsers



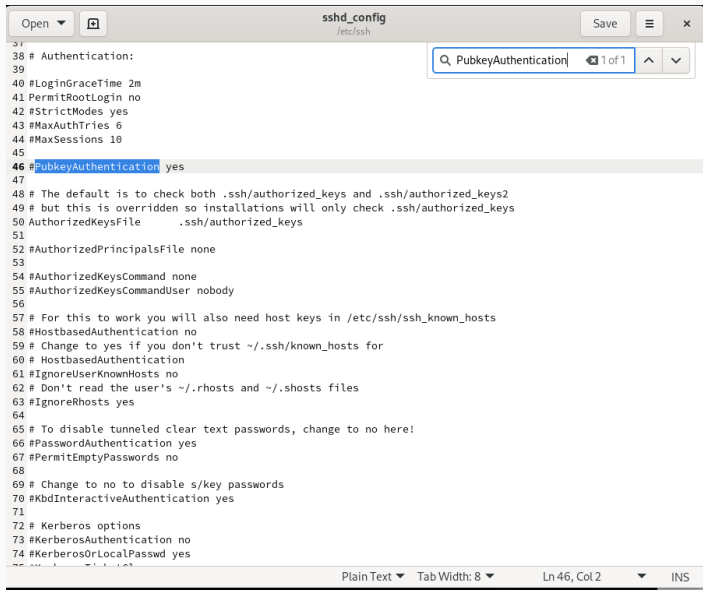
```
root@server:~  
root@server:~ x root@server:~  
(gedit:6861): dconf-WARNING **: 16:57:30.437: failed to commit changes t  
to execute child process "dbus-launch" (No such file or directory)  
[root@server ~]# systemctl restart sshd  
[root@server ~]# systemctl status -l sshd  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; pres  
   Active: active (running) since Wed 2026-03-11 16:57:35 UTC; 8s ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
   Main PID: 6902 (sshd)  
     Tasks: 1 (limit: 22435)  
    Memory: 1.4M  
       CPU: 28ms  
   CGroup: /system.slice/ssh.service  
           └─6902 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 star
```

Рис. 7: Отказ пользователю вне списка AllowUsers

Разрешение доступа двум пользователям

```
root@server:~  
root@server:~  
(gedit:6861): dconf-WARNING **: 16:57:30.437: failed to commit changes to dconf: Failed  
to execute child process "dbus-launch" (No such file or directory)  
[root@server ~]# systemctl restart sshd  
[root@server ~]# systemctl status -l sshd  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: ena  
   Active: active (running) since Wed 2026-03-11 16:57:35 UTC; 8s ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
  Main PID: 6902 (sshd)  
    Tasks: 1 (limit: 22435)  
   Memory: 1.4M  
      CPU: 28ms  
   CGroup: /system.slice/ssh.service  
           └─6902 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Mar 11 16:57:35 server.user.net systemd[1]: Starting OpenSSH server daemon...  
Mar 11 16:57:35 server.user.net sshd[6902]: error: Bind to port 2022 on 0.0.0.0  
Mar 11 16:57:35 server.user.net sshd[6902]: error: Bind to port 2022 on :: fail  
Mar 11 16:57:35 server.user.net sshd[6902]: Server listening on 0.0.0.0 port 22.  
Mar 11 16:57:35 server.user.net sshd[6902]: Server listening on :: port 22.  
Mar 11 16:57:35 server.user.net systemd[1]: Started OpenSSH server daemon.  
lines 1-18/18 (END)
```

Настройка двух портов para SSH

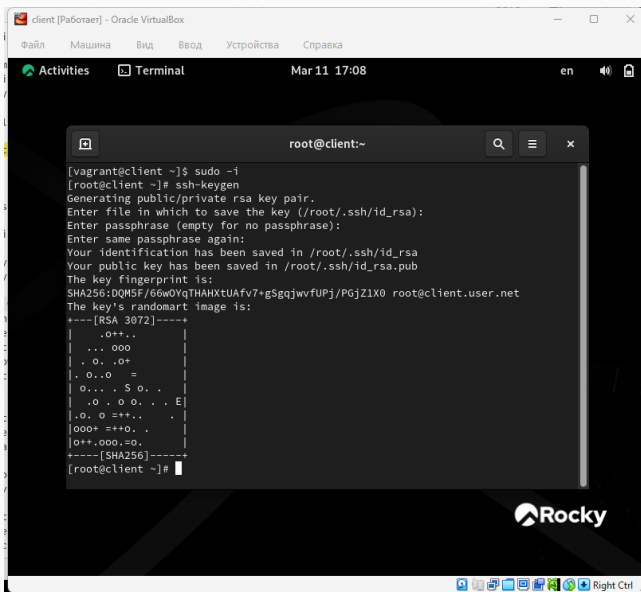


The image shows a text editor window titled 'sshd_config' with the file path '/etc/ssh'. The editor contains the SSH daemon configuration file. A search bar at the top right shows the query 'PubkeyAuthentication' with a result count of '1 of 1'. The configuration file content is as follows:

```
37
38 # Authentication:
39
40 #LoginGraceTime 2m
41 PermitRootLogin no
42 #StrictModes yes
43 #MaxAuthTries 6
44 #MaxSessions 10
45
46 #PubkeyAuthentication yes
47
48 # The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
49 # but this is overridden so installations will only check .ssh/authorized_keys
50 AuthorizedKeysFile      .ssh/authorized_keys
51
52 #AuthorizedPrincipalsFile none
53
54 #AuthorizedKeysCommand none
55 #AuthorizedKeysCommandUser nobody
56
57 # For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
58 #HostbasedAuthentication no
59 # Change to yes if you don't trust ~/.ssh/known_hosts for
60 # HostbasedAuthentication
61 #IgnoreUserKnownHosts no
62 # Don't read the user's ~/.rhosts and ~/.shosts files
63 #IgnoreRhosts yes
64
65 # To disable tunneled clear text passwords, change to no here!
66 #PasswordAuthentication yes
67 #PermitEmptyPasswords no
68
69 # Change to no to disable s/key passwords
70 #KbdInteractiveAuthentication yes
71
72 # Kerberos options
73 #KerberosAuthentication no
74 #KerberosOrLocalPasswd yes
```

The status bar at the bottom indicates 'Plain Text', 'Tab Width: 8', 'Ln 46, Col 2', and 'INS'.

Ошибка bind по порт 2022



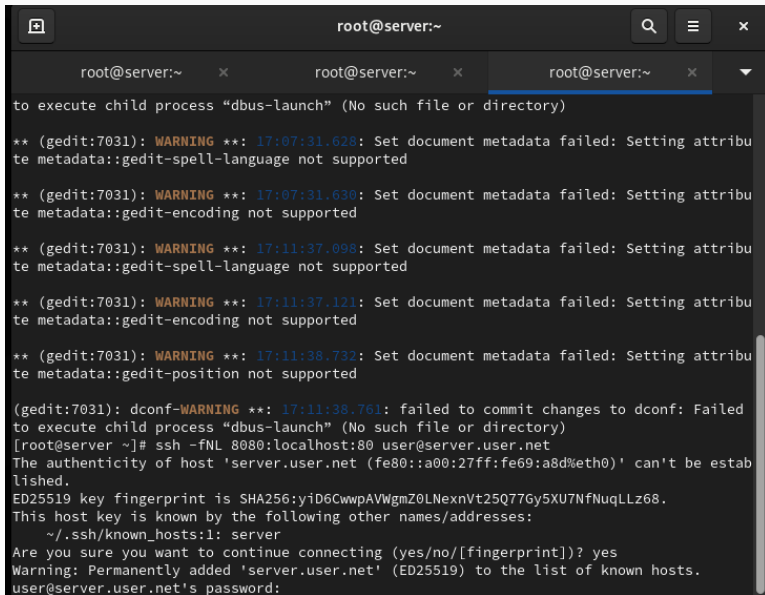
The screenshot shows a terminal window titled "client [Работает] - Oracle VirtualBox". The terminal is running as root on a machine named "client". The user "vagrant" has executed the command "sudo -i" to become root. The root user then runs "ssh-keygen" to generate an SSH key pair. The process involves entering a file path (defaulting to "/root/.ssh/id_rsa"), a passphrase, and confirming the passphrase. The terminal displays the key fingerprint and the randomart image for the generated key.

```
root@client:~  
[vagrant@client ~]$ sudo -i  
[root@client ~]# ssh-keygen  
Generating public/private rsa key pair.  
Enter file in which to save the key (/root/.ssh/id_rsa):  
Enter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Your identification has been saved in /root/.ssh/id_rsa  
Your public key has been saved in /root/.ssh/id_rsa.pub  
The key fingerprint is:  
SHA256:DQMSF/66w0YqTHAHxtUAFv7+gSgqjwvfUPj/PgJZ1X0 root@client.user.net  
The key's randomart image is:  
+---[RSA 3072]-----+  
| .o+.. |  
| ... ooo |  
| . o. .o+ |  
| . o..o = |  
| o... . S o. . |  
| .o . o o. . . E |  
|.o. o =+.. . |  
|ooo+ =+o. . |  
|o++..ooo.=o. |  
+---[SHA256]-----+  
[root@client ~]#
```

Исправление SELinux e firewall para 2022

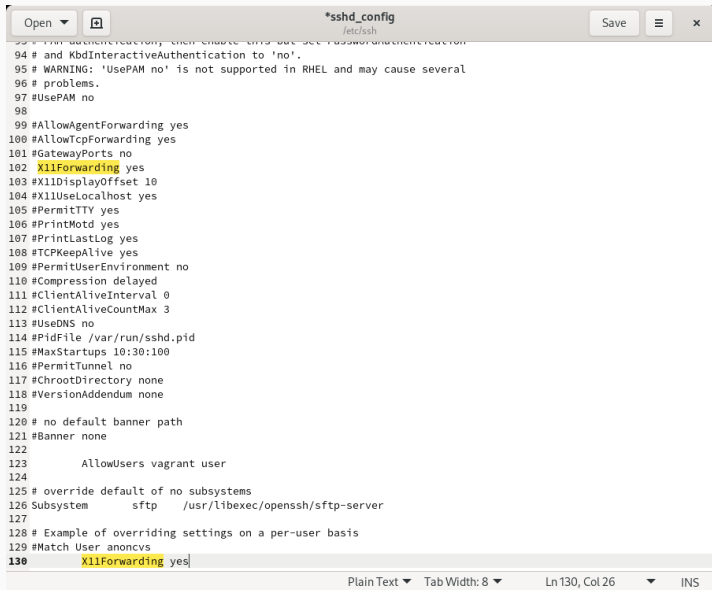
```
vagrant@client:~  
The key's randomart image is:  
+---[RSA 3072]-----+  
|      .o++..      |  
|    ... 000      |  
|  . o. .o+      |  
|. o..o   =      |  
| o... . S o. .  |  
|  .o . o o. . . E|  
|.o. o =++..    .|  
|ooo+ =++o. .   |  
|o++..ooo.=o.   |  
+-----[SHA256]-----+  
[root@client ~]# ssh-copy-id user@server.user.net  
/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"  
/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out  
any that are already installed  
  
/bin/ssh-copy-id: ERROR: ssh: Could not resolve hostname server.user.net: Name o  
r service not known  
  
[root@client ~]#  
logout  
[vagrant@client ~]$
```

Подключение por SSH через порт 2022

A terminal window titled 'root@server:~' with three tabs. The active tab shows the output of a command. The output includes several warnings from gedit about metadata and a successful SSH connection to 'server.user.net' on port 80. The connection is established, and the user is prompted for a password.

```
root@server:~  
to execute child process "dbus-launch" (No such file or directory)  
  
** (gedit:7031): WARNING **: 17:07:31.628: Set document metadata failed: Setting attribute metadata::gedit-spell-language not supported  
  
** (gedit:7031): WARNING **: 17:07:31.630: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported  
  
** (gedit:7031): WARNING **: 17:11:37.098: Set document metadata failed: Setting attribute metadata::gedit-spell-language not supported  
  
** (gedit:7031): WARNING **: 17:11:37.121: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported  
  
** (gedit:7031): WARNING **: 17:11:38.732: Set document metadata failed: Setting attribute metadata::gedit-position not supported  
  
(gedit:7031): dconf-WARNING **: 17:11:38.761: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)  
[root@server ~]# ssh -fNL 8080:localhost:80 user@server.user.net  
The authenticity of host 'server.user.net (fe80::a00:27ff:fe69:a8d%eth0)' can't be established.  
ED25519 key fingerprint is SHA256:yiD6CwwpAVWgmZ0LNexnVt25Q77Gy5XU7NfNuqLLz68.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: server  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'server.user.net' (ED25519) to the list of known hosts.  
user@server.user.net's password:
```


Разрешение аутентификации по ключу



```
Open [icon] *sshd_config [Save] [Menu] [Close]
/etc/ssh
93 # For a more detailed discussion, see the OpenSSH manual page.
94 # and KbdInteractiveAuthentication to 'no'.
95 # WARNING: 'UsePAM no' is not supported in RHEL and may cause several
96 # problems.
97 #UsePAM no
98
99 #AllowAgentForwarding yes
100 #AllowTcpForwarding yes
101 #GatewayPorts no
102 X11Forwarding yes
103 #X11DisplayOffset 10
104 #X11UseLocalhost yes
105 #PermitTTY yes
106 #PrintMotd yes
107 #PrintLastLog yes
108 #TCPKeepAlive yes
109 #PermitUserEnvironment no
110 #Compression delayed
111 #ClientAliveInterval 0
112 #ClientAliveCountMax 3
113 #UseDNS no
114 #PidFile /var/run/sshd.pid
115 #MaxStartups 10:30:100
116 #PermitTunnel no
117 #ChrootDirectory none
118 #VersionAddendum none
119
120 # no default banner path
121 #Banner none
122
123         AllowUsers vagrant user
124
125 # override default of no subsystems
126 Subsystem      sftp        /usr/libexec/openssh/sftp-server
127
128 # Example of overriding settings on a per-user basis
129 #Match User anoncvs
130 X11Forwarding yes
```

Plain Text ▾ Tab Width: 8 ▾ Ln 130, Col 26 ▾ INS

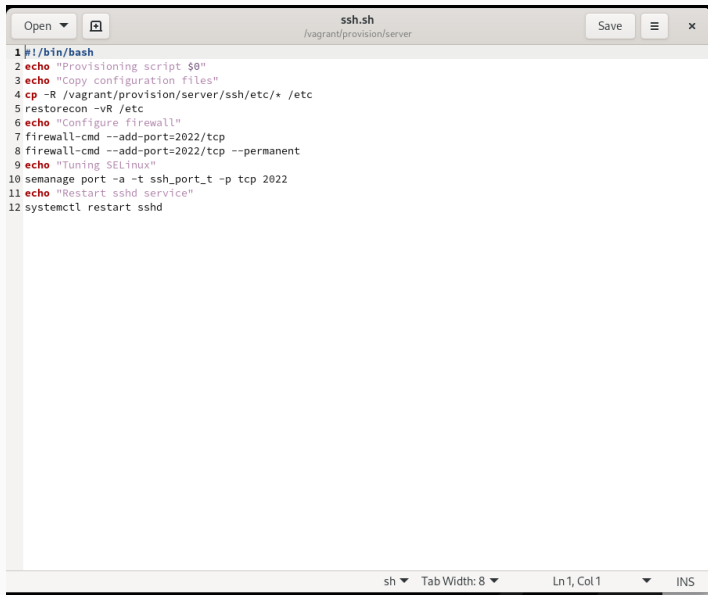
Подключение por ключу sem senha

```
user@server.user.net's password:
user@server.user.net: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).
[root@server ~]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/ssh/etc/ssh
[root@server server]# cp -R /etc/ssh/sshd_config /vagrant/provision/server/ssh/etc/ssh/
[root@server server]# cd /vagrant/provision/server
[root@server server]# touch ssh.sh
[root@server server]# chmod +x ssh.sh
[root@server server]# gedit ssh.sh

(gedit:7114): dconf-WARNING **: 17:17:30.664: failed to commit changes to dconf: Failed
to execute child process "dconf-launch" (No such file or directory)
```

Рис. 14: Подключение por ключу

SSH-туннель и перенаправление портов



The image shows a terminal window titled 'ssh.sh' with the path '/vagrant/provision/server'. The window contains a shell script with 12 lines of code. The script starts with a shebang and proceeds to set up an SSH environment, including copying configuration files, configuring the firewall, tuning SELinux, and restarting the SSH service.

```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Copy configuration files"
4 cp -R /vagrant/provision/server/ssh/etc/* /etc
5 restorecon -vR /etc
6 echo "Configure firewall"
7 firewall-cmd --add-port=2022/tcp
8 firewall-cmd --add-port=2022/tcp --permanent
9 echo "Tuning SELinux"
10 semanage port -a -t ssh_port_t -p tcp 2022
11 echo "Restart sshd service"
12 systemctl restart sshd
```

At the bottom of the terminal window, the status bar shows 'sh', 'Tab Width: 8', 'Ln 1, Col 1', and 'INS'.

Доступ к веб-странице через туннель

Рис. 16: Доступ к веб-странице через туннель

Удалённый запуск консольных команд

Рис. 17: Удалённый запуск консольных команд

Просмотр почты по сервере

Рис. 18: Просмотр почты по сервере

X11Forwarding yes

Рис. 19: X11Forwarding yes

Ошибка DISPLAY при X11 Forwarding

Рис. 20: Ошибка DISPLAY при X11 Forwarding

Выводы по проделанной работе

В ходе работы настроены механизмы безопасного удалённого доступа `por SSH`:

запрет входа `para root`, ограничение пользователей через `AllowUsers`, работа через два порта с настройкой `SELinux` и `firewall`, аутентификация `por` ключу, организация `SSH-туннелей` и удалённый запуск приложений.

Проверки подтвердили корректность конфигурации и соблюдение принципов безопасного доступа.